

Genome-wide identification of genetic requirements of *Pseudomonas aeruginosa* PAO1 for rat cardiomyocyte (H9C2) infection by insertion sequencing

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ABSTRACT

Pseudomonas aeruginosa is a major infectious agent among Gram-negative bacteria, which causes both acute and chronic infections. Infections due to *P. aeruginosa* are hard to treat, as it entails various strategies like virulence factors synthesis, drug efflux systems & resistance and protein secretion systems during pathogenesis. Despite extensive research in *Pseudomonas* pathogenesis, novel drug targets and potential therapeutic strategies are urgently needed. In this study, we investigated the genetic requirements of *P. aeruginosa* PAO1 for rat cardiomyocyte (H9C2) infection by insertion sequencing (INSeq). A mutant library comprising ~70,000 mutants of PAO1 was generated and the differentiated form of H9C2 cells (d-H9C2) was infected with the library. The infected d-H9C2 cells were maintained with antibiotic-protection and without any antibiotics in the growth media for 24 h. Subsequently, DNA library for INSeq was prepared, sequenced and fitness analysis was performed. One hundred and thirteen mutants were negatively selected in the infection condition with antibiotic-protection, whereas 143 mutants were negatively selected in antibiotic-free condition. Surprisingly, a higher number of mutants showed enriched fitness than the mutants of reduced fitness during the infection. We demonstrated that the genes associated with flagella and T3SS are important for adhesion and invasion of cardiomyocytes, while pili and proteases are conditionally essential during host cell lysis. Hence, our findings highlight the essential genes for cardiomyocyte infection, particularly during the intracellular phase. The aerotaxis receptor Aer, plays a critical role during intracellular life. Genes such as *flgE*, *flgF*, *flhA*, *flhB*, *flhC*, *flhD*, *flhE*, *flhF*, *flhG*, *flhH*, *flhI*, *flhJ*, *flhK*, *flhL*, *flhM*, *flhN*, *flhO*, *flhP*, *flhQ*, *flhR*, *flhS*, *flhT*, *flhU*, *flhV*, *flhW*, *flhX*, *flhY*, *flhZ*, *flhA*, *flhB*, *flhC*, *flhD*, *flhE*, *flhF*, *flhG*, *flhH*, *flhI*, *flhJ*, *flhK*, *flhL*, *flhM*, *flhN*, *flhO*, *flhP*, *flhQ*, *flhR*, *flhS*, *flhT*, *flhU*, *flhV*, *flhW*, *flhX*, *flhY*, *flhZ*, *flhA*, *flhB*, *flhC*, *flhD*, *flhE*, *flhF*, *flhG*, *flhH*, *flhI*, *flhJ*, *flhK*, *flhL*, *flhM*, *flhN*, *flhO*, *flhP*, *flhQ*, *flhR*, *flhS*, *flhT*, *flhU*, *flhV*, *flhW*, *flhX*, *flhY*, *flhZ*, *flhA*, *flhB*, *flhC*, *flhD*, *flhE*, *flhF*, *flhG*, *flhH*, *flhI*, *flhJ*, *flhK*, *flhL*, *flhM*, *flhN*, *flhO*, *flhP*, *flhQ*, *flhR*, *flhS*, *flhT*, *flhU*, *flhV*, *flhW*, *flhX*, *flhY*, *flhZ*, *flhA*, *flhB*, *flhC*, *flhD*, *flhE*, *flhF*, *flhG*, *flhH*, *flhI*, *flhJ*, *flhK*, *flhL*, *flhM*, *flhN*, *flhO*, *flhP*, *flhQ*, *flhR*, *flhS*, *flhT*, *flhU*, *flhV*, *flhW*, *flhX*, *flhY*, *flhZ*, *flhA*, *flhB*, *flhC*, *flhD*, *flhE*, *flhF*, *flhG*, *flhH*, *flhI*, *flhJ*, *flhK*, *flhL*, *flhM*, *flhN*, *flhO*, *flhP*, *flhQ*, *flhR*, *flhS*, *flhT*, *flhU*, *flhV*, *flhW*, *flhX*, *flhY*, *flhZ*, *flhA*, *flhB*, *flhC*, *flhD*, *flhE*, *flhF*, *flhG*, *flhH*, *flhI*, *flhJ*, *flhK*, *flhL*, *flhM*, *flhN*, *flhO*, *flhP*, *flhQ*, *flhR*, *flhS*, *flhT*, *flhU*, *flhV*, *flhW*, *flhX*, *flhY*, *flhZ*, *flhA*, *flhB*, *flhC*, *flhD*, *flhE*, *flhF*, *flhG*, *flhH*, *flhI*, *flhJ*, *flhK*, *flhL*, *flhM*, *flhN*, *flhO*, *flhP*, *flhQ*, *flhR*, *flhS*, *flhT*, *flhU*, *flhV*, *flhW*, *flhX*, *flhY*, *flhZ*, *flhA*, *flhB*, *flhC*, *flhD*, *flhE*, *flhF*, *flhG*, *flhH*, *flhI*, *flhJ*, *flhK*, *flhL*, *flhM*, *flhN*, *flhO*, *flhP*, *flhQ*, *flhR*, *flhS*, *flhT*, *flhU*, *flhV*, *flhW*, *flhX*, *flhY*, *flhZ*, *flhA*, *flhB*, *flhC*, *flhD*, *flhE*, *flhF*, *flhG*, *flhH*, *flhI*, *flhJ*, *flhK*, *flhL*, *flhM*, *flhN*, *flhO*, *flhP*, *flhQ*, *flhR*, *flhS*, *flhT*, *flhU*, *flhV*, *flhW*, *flhX*, *flhY*, *flhZ*, *flhA*, *flhB*, *flhC*, *flhD*, *flhE*, *flhF*, *flhG*, *flhH*, *flhI*, *flhJ*, *flhK*, *flhL*, *flhM*, *flhN*, *flhO*, *flhP*, *flhQ*, *flhR*, *flhS*, *flhT*, *flhU*, *flhV*, *flhW*, *flhX*, *flhY*, *flhZ*, *flhA*, *flhB*, *flhC*, *flhD*, *flhE*, *flhF*, *flhG*, *flhH*, *flhI*, *flhJ*, *flhK*, *flhL*, *flhM*, *flhN*, *flhO*, *flhP*, *flhQ*, *flhR*, *flhS*, *flhT*, *flhU*, *flhV*, *flhW*, *flhX*, *flhY*, *flhZ*, *flhA*, *flhB*, *flhC*, *flhD*, *flhE*, *flhF*, *flhG*, *flhH*, *flhI*, *flhJ*, *flhK*, *flhL*, *flhM*, *flhN*, *flhO*, *flhP*, *flhQ*, *flhR*, *flhS*, *flhT*, *flhU*, *flhV*, *flhW*, *flhX*, *flhY*, *flhZ*, *flhA*, *flhB*, *flhC*, *flhD*, *flhE*, *flhF*, *flhG*, *flhH*, *flhI*, *flhJ*, *flhK*, *flhL*, *flhM*, *flhN*, *flhO*, *flhP*, *flhQ*, *flhR*, *flhS*, *flhT*, *flhU*, *flhV*, *flhW*, *flhX*, *flhY*, *flhZ*, *flhA*, *flhB*, *flhC*, *flhD*, *flhE*, *flhF*, *flhG*, 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shown that the *P. aeruginosa* infection induces apoptosis in rat cardiomyocytes (Ranjani et al., 2015). In the present study, we analyzed the conditional essentiality of the *P. aeruginosa* genes during cardiomyocyte infection by employing insertion sequencing (INSeq).

INSeq is a reliable technique, which is a combination of mutagenesis and high throughput sequencing to screen a larger number of mutants with the relatively shortest duration. It is a robust and powerful technique for the rapid correlation of genotype to phenotype in a wide range of bacterial species. Based on transposon mutagenesis, various high throughput sequencing methods such as TraDIS, HITS, Tn-Seq, Tn-Seq Circle and INSeq have been developed. A well-established protocol for transposon library-based sequencing has been reported. (Goodman et al., 2011; van Opijnen et al., 2015). Mutagenesis based sequencing has been performed in many bacterial species particularly to investigate the conditionally essential genes during host-pathogen interactions (Lee et al., 2015; Gallagher et al., 2015; Klein et al., 2015; Shan et al., 2015; Verhagen et al., 2014; de Vries et al., 2014; Goodman et al., 2009; Le Breton et al., 2015). For instance, Turner et al. (2014) demonstrated the genetic requirements of *P. aeruginosa* for acute burn wound and chronic infection by combining the RNA sequencing and insertion sequencing in a murine model. They found that the mutant fitness is not correlated with the expression of a gene in vivo except for some metabolic genes. Chemotactic motility and virulence factors are essential for acute burn wound infection but not for chronic wound infections. They concluded that the fitness of *P. aeruginosa* during infections is determined by the host physiology like the immune system (Turner et al., 2014). The same group investigated the essential genes of the strains PAO1 and PA14 for their survival in cystic fibrosis sputum. Genes involved in the biosynthesis of biotin, riboflavin, pantothenate, purines, and pyrimidines are essential to survive in CF sputum. The essentiality of the genes differed between the two strains (Turner et al., 2015). Skurnik et al. (2013) reported the conditionally essential genes in *P. aeruginosa* PA14 during mucosal colonization and systemic dissemination by INSeq. Mutants associated with several virulence factors such as pyocyanin, LPS, secretion systems, and quorum sensing were deficient in colonization and systemic dissemination. Whereas genes implicated in transport of small molecules, motility, and chemotaxis showed enhanced fitness during cecal colonization but not in dissemination. Poulsen et al. (2019) defined a core genome of *P. aeruginosa* and proposed novel drug targets. They identified 321 core essential genes from clinical isolates grown in analogous growth conditions such as serum, sputum and urine.

The aim of this study is to unveil the genetic requirement of *P. aeruginosa* PAO1 to infect the rat cardiomyocytes through high throughput insertion sequencing. Consequently, we investigated the fitness of PAO1 mutants in two different conditions such as in-presence and absence of amikacin in cardiomyocyte growth media. Our data substantiate the fact that the flagella and T3SS are conditionally essential for adhesion and invasion whereas the pili and proteases are essential for host cell lysis.

2. Materials and methods

2.1. Bacterial strains and plasmids

P. aeruginosa PAO1 was routinely cultured in Luria-Bertani (LB) broth at 37 °C. Bacteria at exponential phase were used for infecting the host cells. *E. coli* Sm10λpir harboring pSAM_BT20 plasmid was maintained in LB broth with 100 µg/ml ampicillin. PAO1 mutant strains were procured from *P. aeruginosa* mutant library (Table S1), University of Washington and maintained in LB with 5 µg/ml tetracycline (Jacobs et al., 2003; Held et al., 2012).

2.2. Cell line maintenance and differentiation

H9C2 cell line was obtained from the National Centre for Cell Science, Pune, India. Cells were cultured in Dulbecco's Modified Eagle's

Medium (DMEM) F12 Ham supplemented with 10% Fetal Bovine Serum (FBS), 100 U/ml of penicillin, 100 µg/ml of streptomycin and 2.5 µg/ml of amphotericin B in a humidified incubator with 5% CO₂ at 37 °C. Cardiomyoblast differentiation was induced as previously described (Ranjani et al., 2015). Briefly, H9C2 cells were seeded in 75 cm² tissue culture flasks at a density of 3.5×10^4 cells in the growth medium. At 50–60% of confluence, high serum media was replaced by low serum media (1% FBS) and incubated for 4 days in the dark with a daily supplement of 1 µM all-trans retinoic acid. On the fifth day of treatment, differentiated cells were trypsinized with 0.025% trypsin-EDTA solution and seeded for further experiments.

2.3. Generation of PAO1 mutant library

P. aeruginosa PAO1 mutant library was generated according to Skurnik et al. (2013) with few modifications. Briefly, an overnight culture of *E. coli* Sm10λpir pSAM_BT20 (Sivakumar et al., 2019) and *P. aeruginosa* PAO1 were mixed and centrifuged. The pellet was washed in LB, resuspended in 100 µl of LB, spotted on pre-warmed LB agar plates and incubated at 37 °C for 3 h. The conjugation spots were suspended in LB and plated on LB agar containing 25 µg/ml of irgasan and 40 µg/ml of gentamicin. After 16 h of incubation, the colonies were collected with a cell scraper, suspended in PBS with 15% glycerol and stored at –80 °C until further use. Approximately 70,000 mutants were generated and the transposition was confirmed by amplifying the gentamicin cassette and transposase using the primers GENF-GENR and TRANSF-TRANSR respectively (Table S1).

2.4. d-H9C2 infection with mutant library

Infection assays were performed according to our previous study (Ranjani et al., 2015). d-H9C2 cells were infected with the mutant library at 1:25 MOI for 1 h. After the infection period, the d-H9C2 cells were incubated with amikacin containing growth media for 1 h. Further, the cells were washed and maintained in two different conditions such as in-presence and absence of amikacin in the growth media. At 24 h of post-infection, infected cells were harvested, and genomic DNA was isolated (Qiagen DNeasy blood and tissue kit).

2.5. Library preparation for insertion sequencing

DNA library was prepared as previously mentioned (Sivakumar et al., 2019). Briefly, insertion junctions were amplified using BiosamaA primer followed by clean up and streptavidin bead binding. The second strand was synthesized using random hexamers (Roche, USA) followed by *MmeI* digestion. After digestion, adapters were ligated, amplified and size selected. Sequencing was performed in Ion Torrent personal genome machine using a 318v2 chip and a 200 bp sequencing chemistry.

2.6. Data analysis

Adapters were trimmed from the raw reads using Cutadapt v1.14, and the reads were split based on barcodes using FastX v2.3 barcode splitter. The analysis was performed using the online software ESSENTIALS (<http://bamics2.cmbi.ru.nl/websoftware/essentials>) by choosing *P. aeruginosa* PAO1 (NC_002516) as the reference genome. Briefly, essential genes were identified based on the FDR adjusted *p*-value of the replicates (<0.05), and the ratio of the expected number of reads, calculated from the number of possible insertion sites per gene, was lower than the fold change cut-off predicted by ESSENTIALS. The fitness of mutants was determined based on their fold changes. Mutants were considered as negatively selected/ reduced fitness, when they had a 2-fold ($\log_2 < -1$, *p*-value <0.05) lower number of read counts per gene in the test condition compared to the control and vice versa. The negative binomial distribution statistical model in EdgeR was used in ESSENTIALS and the dispersion in the data was estimated with quantile-

adjusted conditional maximum likelihood (qCML) (Zomer et al., 2012).

2.7. Live/dead cell assay

The cytotoxic effect of the wild-type and mutant strains were analyzed by live/dead cell assay using the High Content Screening (HCS) System (Operetta, Perkin Elmer, USA). d-H9C2 cells were infected with wild-type and mutants for 1 h followed by incubation in amikacin-containing growth media. At 12 h of post-infection, media was aspirated carefully, and cells were stained with Hoechst 33342 and PI (Invitrogen, USA). Images were captured, and analysis was performed in Harmony software. Twelve fields were analyzed per condition in triplicate, and the results were given as mean values.

2.8. CFU enumeration

Infection was performed as mentioned in the Section 2.7 and the cells were harvested by scrapping at 12 h of post-infection periods and lysed with 0.1% Triton X-100 in PBS. The lysate was plated onto antibiotic-free King's B media after serial dilution.

2.9. Image process and statistical analysis

HCS Images were processed using ImageJ 1.49 (Schneider et al., 2012). All the experiments were performed in triplicates, and the data represented in bar charts are the means of three independent experiments with standard deviation. Statistical analyses were performed using GraphPad QuickCalcs.

3. Results and discussion

3.1. Unique insertions and essential genes

A total of ~7.5 million reads were obtained from the triplicate samples of input library (DMEM), and 24 h infected cells in two different conditions. About 65,536 unique insertions were found in both strands of the genome, which illustrates that an average density of 10.5 insertions/kb (~insertion at every 95 bp) in the mutant library. Based on the density plot, 471 genes were identified as essential for PAO1 to survive in the DMEM medium. Among 471, two hundred and sixty-eight genes were already reported as the core essential genome of *P. aeruginosa* by Poulsen et al. (2019). Previously described essential genes of *P. aeruginosa* (Liberati et al., 2006; Turner et al., 2014; Lee et al., 2015) were compared with our results, including the orthologs of PAO1 for the essential genes reported in PA14 (Skurnik et al., 2013; Poulsen et al., 2019). The orthologs were found from the *Pseudomonas* database (Winsor et al., 2016). The comparison study revealed an intersection of 89 essential genes as the common core set in all these studies (Fig. 1). Genes associated with ribosomal structure, cell wall/membrane/envelope and their biogenesis were found to be the major categories of the essential genes followed by various metabolite transport and metabolism. Although many genes such as *dnaB* involve in DNA replication, *rpoC* encodes RNA polymerase, and the secretion apparatus gene *secE* were identified as essential in the present study, not represented in the core set as reported earlier by Lee et al., 2015. This low concordance between various research outcomes may be due to the inconsistency in sequencing or analytical methods (Martínez-Carranza et al., 2019). However, the essential genes of the pathogen are the potential drug targets, and hence, further exploration of these essential genes,

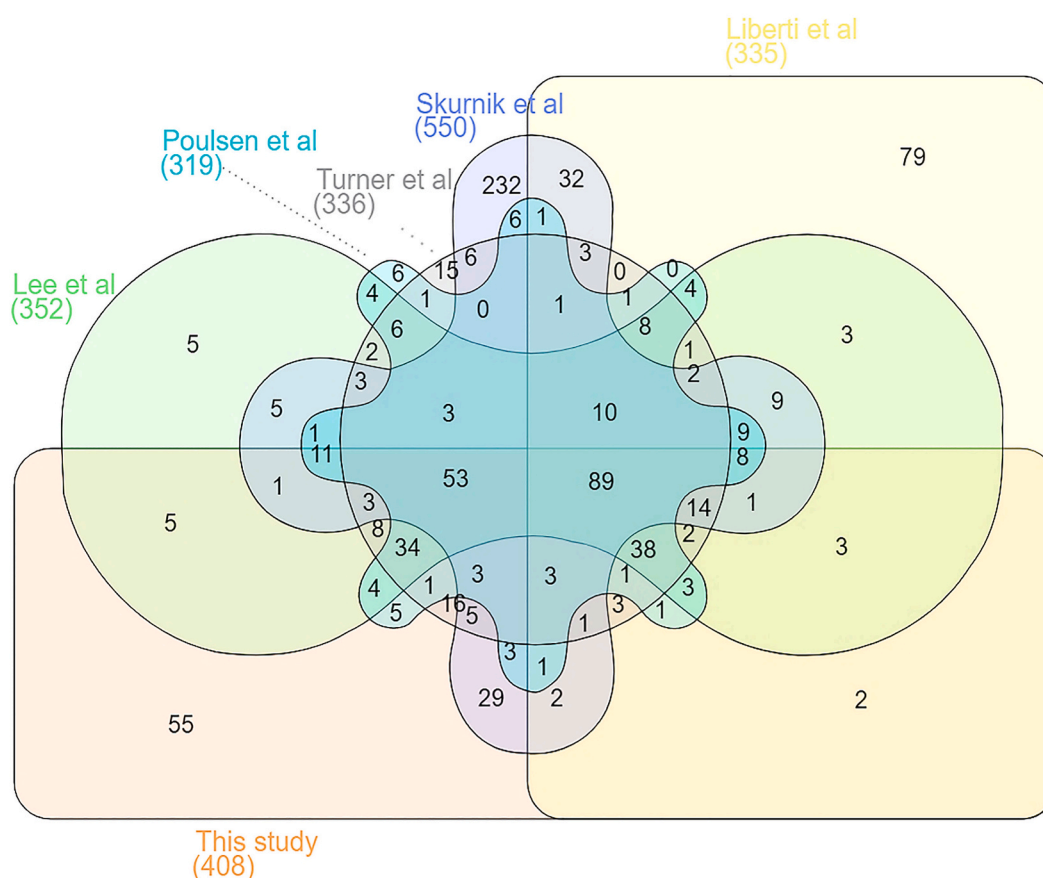


Fig. 1. The intersection of the identified essential genes with previous studies: Among 471 essential genes, 89 genes are identified as essential in all the previous studies (Core-set) and 293 genes were reported already in any one of these studies. Venn diagram was generated using Interactivenn (Heberle et al., 2015).

particularly the core set of essential genes, may lead to promising candidate drug targets. A list of identified essential genes and COG categories is provided in Table S2.

3.2. Genetic requirements of PAO1 for cardiomyocyte infection

Although many studies revealed essential genes of *Pseudomonas* for infection, none of them were at the cellular level, particularly at the intracellular condition. Here we maintained the infected cardiomyocytes with antibiotic-protection and without antibiotics. During antibiotic-protection, pathogens that are actively released from the host as well as during host lysis will be killed by the extracellular antibiotic–amikacin. Since mammalian cells are impermeable to amikacin, internalized pathogens will be protected. On other hand, withdrawal of antibiotics after rigorously washing the infected cells aids the maintenance and growth of the whole population of the pathogen. Hence, post-infection in the presence of antibiotics permits specifically to study the intracellular population and without antibiotics is possible to analyze the entire population including bacteria released from the host cells.

3.2.1. Conditionally essential genes of PAO1 for cardiomyocyte adhesion and infection

Our INSeq analysis identified a higher number of positively selected genes (301) than negatively selected ones (113) in antibiotic-protection condition after infection (Fig. 2a and b, Table S3). Interestingly, genes co-located within operons were over-represented in the output population. For example, out of 14 genes of the *nuo* operon, transposon insertion in 13 genes resulted in enhanced fitness of PAO1 strain during H9C2 infection. Mutation in these genes improved ROS tolerance in *Pseudomonas putida* (Teppo et al., 2018) during host colonization. We speculate similar mechanism could play a role in PAO1 during infection. Inactivation of *nuo* operon genes may provide a selective advantage in tolerating ROS burst, which is the primary defense response of host cells in combating pathogenic bacteria. Earlier, Cole et al. (2017) have reported a higher number of positively selected genes (243) than the negatively selected genes (115) during colonization of *Pseudomonas simiae* WCS417r during *Arabidopsis* root colonization by employing RB-TnSeq analysis. Similarly, Schinner et al. (2020) reported more positively selected genes (112, 255 in *P. aeruginosa* PA14, PAO1,

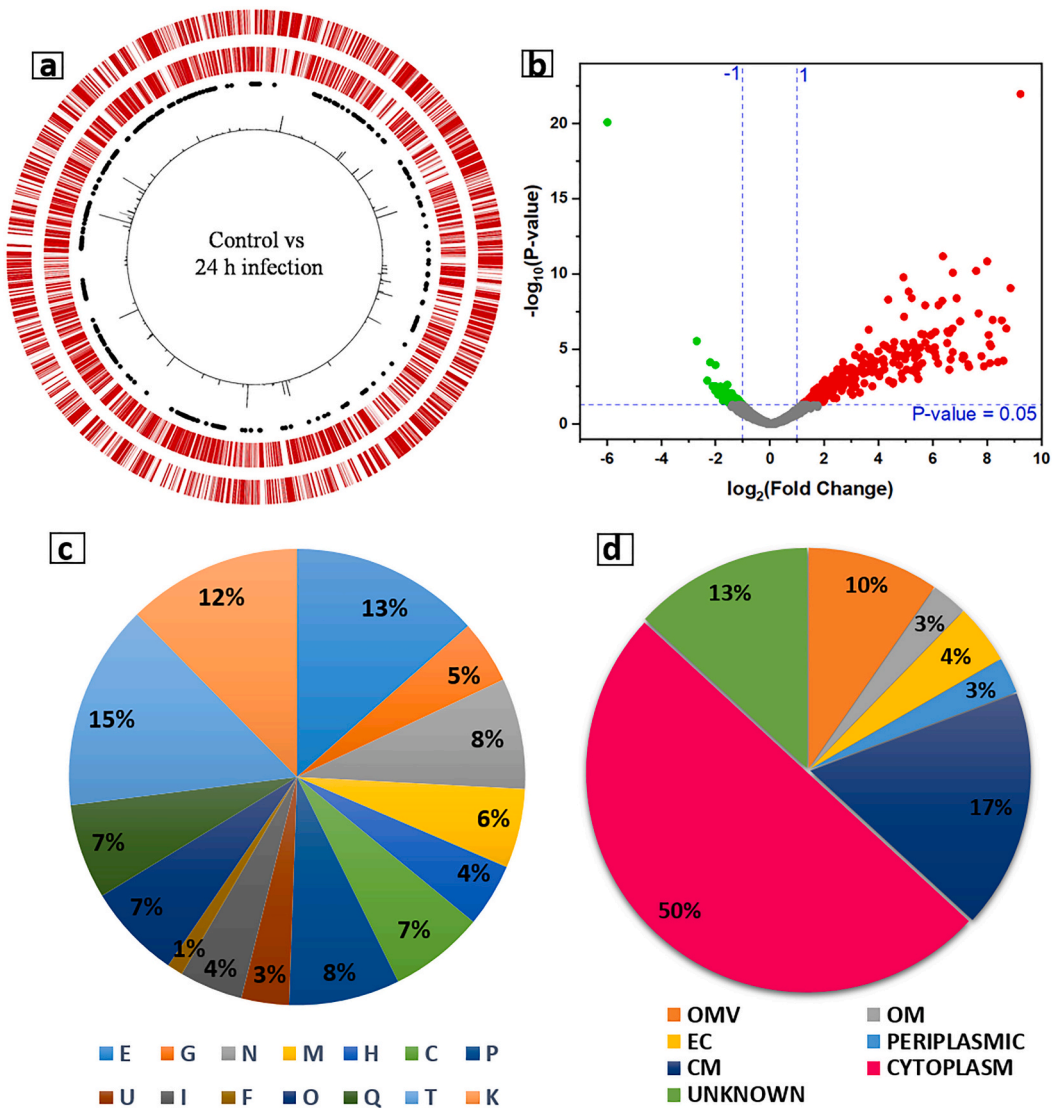


Fig. 2. a. Representation of the fold change occurrence in the mutants during infection condition with amikacin. From the outermost ring to innermost ring: genes on the positive strand, genes on the negative strand, 471 identified essential genes, fold change occurrence in the mutants (outward line-positive fold change, inward line-negative fold change). The circular map was drawn using CiVi, a web-based tool (Overmars et al., 2015). b. Volcano plot representing the fitness of the mutants. c. COG categories of conditionally essential genes for d-H9C2 infection. d. Categories of conditionally essential genes based on localization. Percentage of distribution in a various subcellular location: OMV: Outer membrane vesicles, OM: Outer membrane, EC: Extracellular, CM: Cytoplasmic membrane.

respectively) versus negatively selected genes (45, 49, respectively) during biofilm formation. Among negatively selected mutants, PA1561, PA5205, PA0928, PA5535, PA2197, PA0271, PA2559 and PA5045 showed more than four-fold changes; particularly mutant PA1561 had –63.381-fold difference. PA1561 encodes aerotaxis receptor Aer, PA5205 is a conserved hypothetical MarC-related protein, PA0928 encodes GacS, PA5535, PA2197, PA2559 and PA0271 are hypothetical proteins, and PA5045 is penicillin-binding protein 1A PonA. The fitness of negatively selected mutants is illustrated in Table 1.

Many conditionally essential (negatively-selected) genes encode poorly characterized proteins. Among 113 conditionally essential genes, forty-one are hypothetical proteins. Besides the hypothetical proteins, the conditionally essential genes mainly belong to amino acid transport and metabolism (13%), signal transduction mechanisms (15%), and transcription (12%). Fig. 2c illustrates the distribution of the conditionally essential genes based on their COG categories. Fifty percent of the population localizes in the cytoplasm. About 17% localize in the cytoplasmic membrane, 10% are outer membrane vesicle, 4% are extracellular, and 3% are outer membrane and periplasm (Fig. 2d). Fifteen percent of the conditionally essential genes are already known virulence factors. Of these virulence factors, many genes are associated with flagella such as *flgE*, *flgF*, *flgL*, *flhA*, *flgA*, PA3350, and *motA* (Bouteiller et al., 2021), *pcrV*, *pscD*, and *pscU* belong to type-three secretion system and a type VI secretion system gene PA2371 – *clpV3* (Filloux, 2011). *gacS*, *lasR*, *wbpJ*, *pvdD* and *aprA* are other virulence factors found to be conditionally essential during infection. The fitness level of flagella and T3SS associated mutants suggests their crucial role in adhesion and infection. The flagellum is a complex organelle, an assembly of more than 20 proteins and more than fifty genes involve in the biosynthesis and function (Haiko and Westerlund-Wikström, 2013). Flagella play a vital role in pathogenesis through their motility and chemotaxis properties. Besides its role in motility, it also acts as the ligand for many eukaryotic receptors such as asialo ganglio-N-tetraosylceramide (asialo-GM1), toll-like receptor 2 (TLR2), etc., (Feldman et al., 1998). The whole flagellum aids a significant role in the adhesion of host cells and acts as an antigen for adaptive immunity (Haiko and Westerlund-Wikström, 2013). The importance of flagellar components in host-pathogen interactions was demonstrated in many studies (Verma et al., 2006; Shen et al., 2017; Jyot et al., 2007; Zhang et al., 2007; Andersen-Nissen et al., 2005). Reinforcing the previous studies, our INSeq data also evidenced the significance of flagella specifically during adhesion and infection of cardiomyocytes.

T3SS is a major virulence factor of *P. aeruginosa*, and it is active during acute infections (Hauser et al., 2002). Toxins or effectors are directly injected into the host cell through T3SS to hijack the cellular pathways with consequent cell death. T3SS is a syringe-like structure that is an assembly of over 20 proteins in three complexes called the basal body, the needle, and the translocon. We found *pcrV*, *pscD* and *pscU* are conditionally essential for cardiomyocyte infection in the presence of amikacin. *pcrV* – low calcium response locus protein V is the needle tip of the T3S translocator and is required for the appropriate assembly of the injector (Sato et al., 2011). *pcrV* also regulates the effector translocation and an important protective antigen against T3SS (Frank et al., 2002; Holder et al., 2001). Hence, *PcrV* is a potential target of the therapeutic antibody (Thaden et al., 2016). *PscD* and *PscU* are the components of the basal body where the former present in the inner membrane and the latter present in the cytoplasmic phase (Perdu et al., 2015). *PscU* is an autoprotease and also a ruler protein that regulates the length of the needle (Bergeron et al., 2016). A functional T3SS is required for the cytotoxic effects on polymorphonuclear neutrophils (PMNs) and to resist the bactericidal effect of PMNs (Dacheux et al., 1999).

Around 37% of conditionally essential gene products belong to membrane proteins including OMVs, outer membrane, periplasmic and cytoplasmic membrane. Ten percent of the products localize in OMV. OMVs are the spherical vesicles bud off from the outer membrane filled

Table 1

Conditionally essential genes of PAO1 in antibiotic-protection condition after d-H9C2 infection.

Locus tag	Fold change	p-value	Gene	Product
OMVs				
PA0782	–2.4396	0.01669	<i>putA</i>	bifunctional proline dehydrogenase/pyrroline-5-carboxylate dehydrogenase
PA0888	–3.2107	0.008	<i>aotJ</i>	arginine/ornithine binding protein AotJ
PA0919	–2.1828	0.03901	–	alanine-phosphatidylglycerol hydrolase
PA0955	–3.1087	0.00807	–	hypothetical protein
PA1080	–3.1333	0.0105	<i>flgE</i>	flagellar hook protein FlgE
PA1288	–2.1854	0.03928	–	probable outer membrane protein precursor
PA1561	–63.381	8.31E-21	<i>aer</i>	aerotaxis receptor Aer
PA1767	–2.7296	0.04315	–	hypothetical protein
PA1791	–2.2336	0.03736	–	hypothetical protein
PA3147	–2.3692	0.0458	<i>wbpJ</i>	glycosyl transferase WbpJ
PA5045	–4.0141	0.00012	<i>ponA</i>	penicillin-binding protein 1A
Outer membrane				
PA0931	–2.2044	0.02539	<i>pirA</i>	ferric enterobactin receptor PirA
PA1288	–2.1854	0.03928	–	probable outer membrane protein precursor
PA1365	–2.6311	0.01503	–	siderophore receptor
Cytoplasmic membrane				
PA0924	–2.0471	0.04254	–	hypothetical protein
PA0928	–4.953	0.00128	<i>gacS</i>	sensor/response regulator hybrid
PA1072	–2.1628	0.03487	<i>braE</i>	branched-chain amino acid transport protein BraE
PA1091	–2.4261	0.01734	<i>fgtA</i>	flagellar glycosyl transferase, FgtA
PA1317	–2.548	0.03385	<i>cyoA</i>	cytochrome o ubiquinol oxidase subunit II
PA1433	–2.2658	0.03999	–	conserved hypothetical protein
PA1519	–2.3636	0.03116	–	probable transporter
PA1647	–3.256	0.00483	–	probable sulfate transporter
PA1690	–2.7199	0.03752	<i>pscU</i>	translocation protein in type III secretion
PA1854	–3.3918	0.00418	–	conserved hypothetical protein
PA1930	–2.6393	0.01228	–	probable chemotaxis transducer
PA1947	–3.3875	0.00359	<i>rbsA</i>	ribose transporter RbsA
PA2041	–2.3289	0.03717	–	amino acid permease
PA2477	–2.8329	0.04039	–	thiol:disulfide interchange protein
PA2563	–2.6857	0.01382	–	probable sulfate transporter
PA2763	–3.6897	0.00865	–	hypothetical protein
PA3210	–2.1075	0.04404	<i>trkH</i>	potassium uptake protein TrkH
PA3258	–2.1125	0.04022	–	hypothetical protein
PA4954	–3.1456	0.01138	<i>motA</i>	chemotaxis protein MotA
PA5205	–6.4716	2.98E-06	–	multiple drug resistance protein MarC
Periplasm				
PA0888	–3.2107	0.008	<i>aotJ</i>	arginine/ornithine binding protein AotJ
PA1081	–3.3764	0.01122	<i>flgF</i>	flagellar basal body rod protein FlgF
PA2592	–2.7762	0.00893	–	periplasmic spermidine/putrescine-binding protein
Extracellular				
PA0888	–3.2107	0.008	<i>aotJ</i>	arginine/ornithine binding protein AotJ
PA1087	–2.6938	0.01524	<i>flgL</i>	flagellar hook-associated protein FlgL
PA1249	–2.1904	0.04672	<i>aprA</i>	alkaline metalloproteinase
PA1706	–3.2059	0.00746	<i>pcrV</i>	type III secretion protein PcrV
PA2592	–2.7762	0.00893	–	periplasmic spermidine/putrescine-binding protein
Cytoplasmic Transcriptional regulators				
PA1097	–2.927	0.00968	<i>fleQ</i>	transcriptional regulator FleQ
PA1430	–2.6187	0.02974	<i>lasR</i>	transcriptional regulator LasR
PA0961	–2.5597	0.03495	–	cold-shock protein

(continued on next page)

Table 1 (continued)

Locus tag	Fold change	p-value	Gene	Product
PA1145	-3.6167	0.00795	–	transcriptional regulator
PA1182	-2.4465	0.04917	–	transcriptional regulator
PA1738	-2.4321	0.032	–	transcriptional regulator
PA1799	-2.9742	0.03052	<i>parR</i>	two-component response regulator, ParR
PA1836	-2.4885	0.02251	–	transcriptional regulator
PA1850	-2.6145	0.01835	–	transcriptional regulator
PA2492	-2.6109	0.01398	<i>mexT</i>	transcriptional regulator MexT
Cytoplasm				
PA1455	-3.1303	0.01171	<i>flhA</i>	flagellar biosynthesis sigma factor
PA1900	-3.0704	0.0191	<i>phzB2</i>	phenazine biosynthesis protein
PA2371	-3.239	0.00927	–	ClpA/B-type protease
PA2399	-2.6716	0.00923	<i>pvdD</i>	pyoverdine synthetase D
PA3350	-3.1675	0.00647	–	flagellar basal body P-ring biosynthesis protein FlgA
PA1717	-3.0024	0.0191	<i>pscD</i>	type III export protein PscD
PA0865	-2.1075	0.04146	<i>hpd</i>	4-hydroxyphenylpyruvate dioxygenase
PA0935	-2.2444	0.04751	–	nucleoside triphosphate pyrophosphohydrolase
PA1001	-2.6129	0.04652	<i>phnA</i>	anthranilate synthase component I
PA1127	-2.6835	0.03552	–	Oxidoreductase
PA1135	-3.2221	0.0048	–	chaperone protein HchA
PA1185	-2.7152	0.0294	–	glutathione S-transferase
PA1254	-2.5238	0.03991	<i>lphC</i>	delta1-pyrroline-4-hydroxy-2-carboxylate deaminase, LphC
PA1524	-2.1231	0.04139	<i>xdhA</i>	xanthine dehydrogenase
PA1562	-2.2613	0.02752	<i>acnA</i>	aconitate hydratase
PA1622	-2.3942	0.03611	–	Hydrolase
PA1668	-2.9373	0.02847	<i>dotU2</i>	DotU2
PA1736	-3.7463	0.01075	–	acetyl-CoA acetyltransferase
PA1737	-2.7015	0.01306	–	3-hydroxyacyl-CoA dehydrogenase
PA1739	-2.6801	0.04158	–	Oxidoreductase
PA1752	-3.3944	0.01072	–	2-dehydropantoate 2-reductase
PA1757	-2.6754	0.02324	<i>thrH</i>	homoserine kinase
PA1770	-2.9749	0.00239	<i>ppsA</i>	phosphoenolpyruvate synthase
PA1813	-3.2049	0.02144	–	hydroxyacylglutathione hydrolase
PA1818	-2.6792	0.00858	<i>ldcA</i>	lysine-specific pyridoxal 5'-phosphate-dependent carboxylase LdcA
PA1823	-2.4956	0.03645	<i>nudC</i>	NADH pyrophosphatase
PA2035	-2.7203	0.0149	–	thiamine pyrophosphate protein
PA2103	-2.3197	0.02425	–	molybdopterin biosynthesis protein MoeB
PA2144	-2.3821	0.04093	<i>glgP</i>	glycogen phosphorylase
PA2188	-2.5295	0.01914	–	probable alcohol dehydrogenase (Zn-dependent)
PA2570	-2.5415	0.04742	<i>lecA</i>	LecA protein
PA3863	-2.3343	0.02434	<i>dauA</i>	FAD-dependent catabolic D-arginine dehydrogenase DauA
PA5535	-4.5992	7.68E-05	–	hypothetical protein
PA0891	-2.5158	0.02319	–	hypothetical protein
PA1076	-2.8645	0.02691	–	hypothetical protein
PA2197	-4.3132	0.00309	–	hypothetical protein
PA2095	-3.333	0.02602	–	hypothetical protein
PA1896	-3.0579	0.0158	–	hypothetical protein
PA1915	-2.3347	0.04875	–	hypothetical protein
PA1305	-2.7817	0.03314	–	hypothetical protein
PA1368	-3.049	0.0107	–	hypothetical protein
PA1374	-2.7241	0.02733	–	hypothetical protein
PA1387	-2.6295	0.0091	–	hypothetical protein
PA0821	-2.2887	0.02052	–	hypothetical protein
PA0644	-2.1454	0.04039	–	hypothetical protein
PA1208	-2.2352	0.04689	–	hypothetical protein
PA1731	-2.5364	0.0399	–	conserved hypothetical protein
Unknown				
PA0271	-4.0496	0.00585	–	hypothetical protein
PA0585	-2.1983	0.03579	–	hypothetical protein

Table 1 (continued)

Locus tag	Fold change	p-value	Gene	Product
PA0989	-2.6157	0.01811	–	hypothetical protein
PA1075	-3.0758	0.02012	–	hypothetical protein
PA1672	-2.2512	0.04393	–	hypothetical protein
PA1733	-3.1568	0.00438	–	hypothetical protein
PA1888	-3.5336	0.00318	–	hypothetical protein
PA1892	-2.3801	0.03486	–	hypothetical protein
PA1894	-2.863	0.00937	–	hypothetical protein
PA2009	-2.9692	0.0095	<i>hmgA</i>	homogentisate 1,2-dioxygenase
PA2238	-3.3017	0.01807	<i>pslH</i>	protein PslH
PA2559	-4.0241	0.00306	–	hypothetical protein
PA2565	-2.5095	0.03674	–	hypothetical protein
PA2854	-2.5873	0.01244	–	hypothetical protein
PA3140	-2.2691	0.03706	–	hypothetical protein

Bold are common negatively selected genes in both conditions of infection.

with periplasmic contents. OMVs involve in various functions like regulating microbial interactions within bacterial communities and with the host (Schwechheimer and Kuehn, 2015). Host cell-targeted delivery, protection from proteolytic dehydration, modulation of the immune response, bacterial effector secretion and distant delivery of proteins are the most significant benefits of OMVs. In a host niche, mostly OMVs are enriched with virulence factors that aid in the host cell invasion, destruction, immune system evasion and antibiotic resistance (Bonington and Kuehn, 2014). PutA, AotJ, FlgE, Aer, WbpJ, and PonA are some gene-products localize in OMVs and the rest are hypothetical proteins. PutA is a bifunctional proline dehydrogenase/pyrroline-5-carboxylate dehydrogenase, AotJ- arginine/ornithine-binding protein, FlgE- flagellar hook protein, Aer- aerotaxis receptor, WbpJ- glycosyl-transferase and PonA is penicillin-binding protein 1A. WbpJ and FlgE have known virulence factors, where the former involve in LPS O-antigen synthesis (Rocchetta et al., 1999) and the latter is a flagellar component. PAO1 utilizes proline as the sole source of carbon and nitrogen by using the bifunctional PutA (Nakada et al., 2002). AotJ belongs to the ABC transporter system which binds to arginine/ornithine. Everett et al. (2017) demonstrated that the virulence of *Pseudomonas* is highly influenced by the presence of arginine in the host environment especially in vivo because arginine is a conditionally essential amino acid for the pathogen. Aer aids the bacterium to identify the specific oxygen and energy-rich niches (Amin et al., 2007). *flgQ*, *lasR*, *mexT*, *parR*, PA0961, PA1145, PA1182, PA1738, PA1799, PA1836, and PA1850 are the conditionally essential transcriptional regulators during this cardiomyocyte infection condition. *FlgQ* is the flagellar transcriptional regulator that highly influences flagellar biogenesis and its related operons (Dasgupta et al., 2002). *FlgQ* also regulates the expression of *pel* genes by acting as both the activator and repressor based on the availability of c-di-GMP (Baraquet et al., 2012). *LasR* is the global regulator of many virulence factors and an important quorum-sensing receptor. Elastases, pyocyanin, and proteases are the virulence factors whose expression is highly regulated by *LasR* (Longo et al., 2013). The role of other conditionally essential transcriptional regulators needs to be analyzed for understanding their potential function during infection.

3.2.2. Conditionally essential genes of PAO1 for cardiomyocyte lysis/active release

We found 143 negatively selected mutants and 128 positively selected mutants, in the antibiotics-free condition after d-H9C2 infection (Fig. 3a and b). About fifteen mutants showed more than four-fold reduced fitness, and nine genes showed more than four-fold increased fitness. Mutants of PA5471, PA0395, PA0271, PA4547, PA1561, PA1101, PA0766, PA1758, PA1374, PA1097, PA4398, PA3824, PA0375, PA4546, and PA0396 showed reduced fitness. Table 2 shows the list of conditionally essential genes and their fold-change during d-H9C2 infection without amikacin and Table S3 is the list of positively

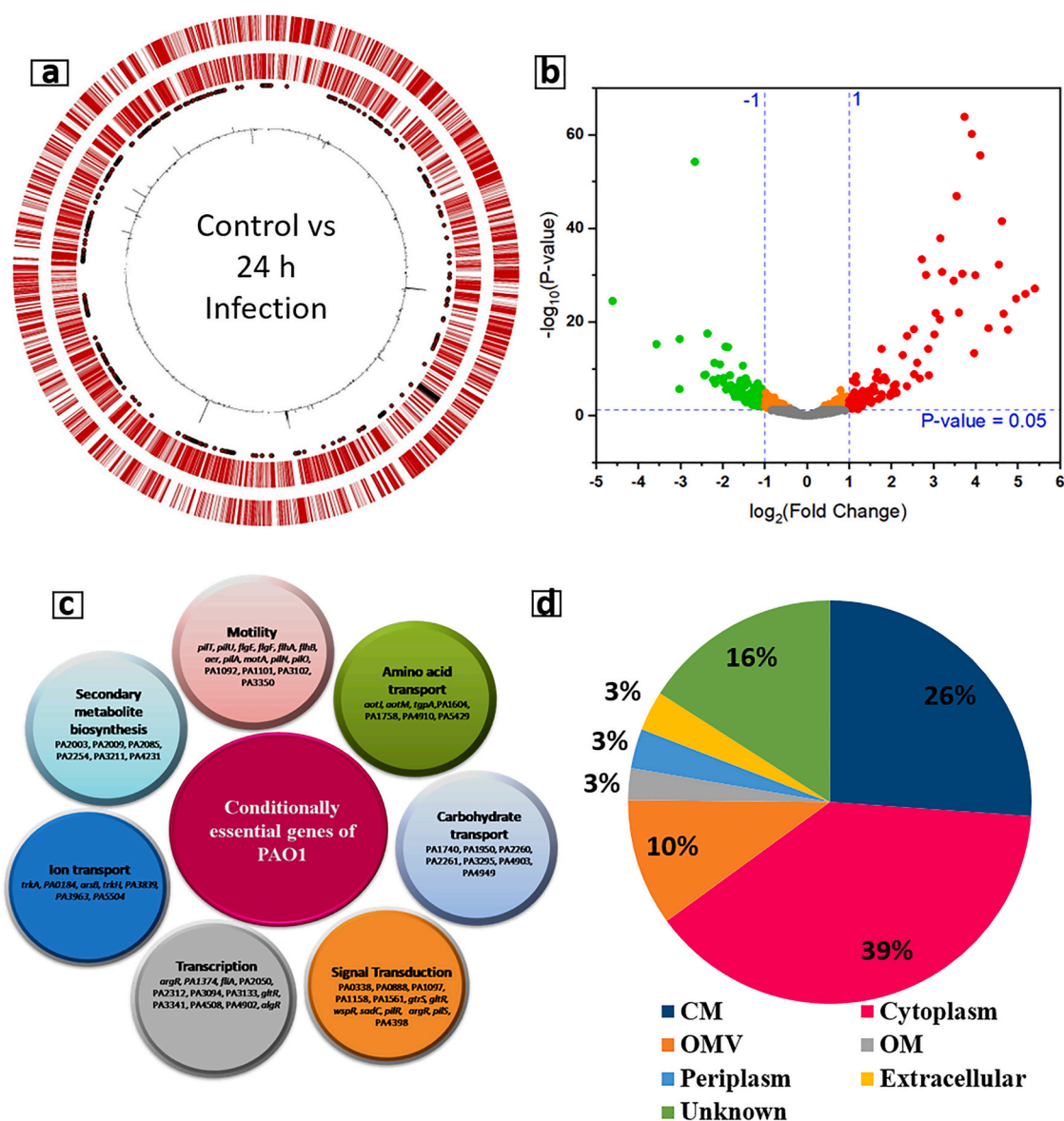


Fig. 3. a. Representation of the fold change occurrence in the mutants in the infection condition without antibiotics. From the outermost ring to innermost ring: genes on the positive strand, genes on the negative strand, 471 identified essential genes, fold change occurrence in the mutants (outward line-positive fold change, inward line-negative fold change). b. Volcano plot representing the fitness of the mutants. c. Representative image of the major COG categories of conditionally essential genes during d-H9C2 infection without amikacin. d. Categories of conditionally essential gene products based on localization. Percentage of distribution in a various subcellular location: OMV: Outer membrane vesicles, OM: Outer membrane, EC: Extracellular, CM: Cytoplasmic membrane.

selected mutants. Many genes involved in motility especially the adhesins such as *pilA*, *pilT*, *pilU*, *pilO*, *pilN*, *flgE*, *flgF*, *flhA*, *flhB*, and *motA* are found to be conditionally essential. Genes associated with signal transduction and transcription are the next major group of conditionally essential genes followed by nutrient transport systems specifically carbohydrate and amino acid (Fig. 3c). Twenty-six percent of conditionally essential gene products for d-H9C2 infection without amikacin are localized in the cytoplasmic membrane. The majority of the gene products belong to the cytoplasm (39%), 10% are outer membrane vesicle, 3% are outer membrane, periplasm and extracellular. Sixteen percent of the population's localization is unknown (Fig. 3d).

The cardiomyocyte infection condition without amikacin retains the entire population including adhered, internalized, released after host lysis, or actively released from the host cells unlike with amikacin which retains only the adhered and internalized. Among 143 conditionally essential genes, nineteen mutants are common in both infection

conditions. Genes associated with flagella such as *flgE*, *flgF*, *flhA*, *flhB*, *flhA*, *fliC*, *fliF*, *motA*, and PA3350 are conditionally essential for cardiomyocyte infection in both conditions. *aotJ*, *aer*, *whpJ*, *ponA*, *fleQ*, PA5205, *hmgA*, *trkH* and *pslH* are the common conditionally essential genes for infection. PA0271, PA1127, PA1374, PA2559 are the other common conditionally essential genes encoding the hypothetical protein. The reason for a few common conditionally essential genes for infection is unclear besides the actuality that the released pathogen can replicate without any constrain in antibiotic-free extracellular conditions. On the contrary, the released pathogen will be killed by amikacin present in the extracellular medium and only the host-protected pathogen will survive and proliferate. Other than the common conditionally essential genes, the crucial categories are the genes involve in type IV pili biosynthesis and regulation such as *pilA*, *pilT*, *pilU*, *pilO*, *pilR*, *pilS* and *pilN* and T3SS such as *pscT*, *pcrD*, *pcrV*, and *exsB*. These data suggest that the flagella and T3SS are essential for adhesion, invasion and host lysis /

Table 2

Conditionally essential genes of PAO1 in antibiotics-free condition after d-H9C2 infection.

Locus Tag	Fold change	p-value	Gene	Product
OMVs				
PA1561	-6.31067	5.79E-55	aer	aerotaxis receptor Aer
PA0766	-5.29562	1.88E-09	<i>mucD</i>	serine protease MucD
PA0375	-4.2858	1.25E-08	<i>ftsX</i>	cell division protein FtsX
PA5045	-3.81166	1.81E-15	ponA	penicillin-binding protein 1A
PA4941	-3.50556	2.91E-07	<i>hflC</i>	protease subunit HflC
PA2235	-3.29461	3.35E-05	<i>pslE</i>	protein PslE
PA3147	-3.28079	2.68E-05	wbpJ	glycosyl transferase WbpJ
PA4639	-3.03123	2.33E-05	–	hypothetical protein
PA4942	-2.80403	3.75E-05	<i>hflK</i>	protease subunit HflK
PA2800	-2.70701	5.96E-05	–	hypothetical protein
PA1080	-2.53511	1.44E-05	flgE	flagellar hook protein FlgE
PA4423	-2.24433	0.000448	–	hypothetical protein
PA4525	-2.23285	0.000179	<i>pilA</i>	type 4 fimbrial PilA
PA0888	-2.12216	0.001585	aotJ	arginine/ornithine binding protein AotJ
PA5258	-2.11919	0.00121	–	hypothetical protein
PA5429	-2.00956	0.001459	<i>aspA</i>	aspartate ammonia-lyase
OM				
PA1101	-5.40447	2.27E-09	<i>flfF</i>	flagellar MS-ring protein
PA1712	-3.19963	3.2E-05	<i>exsB</i>	exoenzyme S synthesis protein B
PA2800	-2.70701	5.96E-05	–	hypothetical protein
PA5107	-2.21454	0.001957	<i>Blc</i>	outer membrane lipoprotein Blc
Cytoplasmic membrane				
PA4398	-4.51676	1.35E-07	–	two-component sensor
PA0375	-4.2858	1.25E-08	<i>ftsX</i>	cell division protein FtsX
PA4546	-4.18337	1.14E-11	<i>pilS</i>	two-component sensor PilS
PA1452	-3.35969	1.25E-05	<i>flhA</i>	flagellar biosynthesis protein FlhA
PA2235	-3.29461	3.35E-05	<i>pslE</i>	protein PslE
PA4910	-3.10339	1.55E-06	–	branched-chain amino acid ABC transporter ATP-binding protein
PA0016	-3.03062	2.57E-07	<i>trkA</i>	potassium transporter peripheral membrane protein
PA2873	-3.00228	4.68E-08	–	hypothetical protein
PA1703	-2.99171	1.06E-05	<i>pcrD</i>	type III secretory apparatus protein PcrD
PA0751	-2.95272	5.24E-06	–	hypothetical protein
PA3210	-2.95042	3.15E-06	trkH	potassium uptake protein TrkH
PA5205	-2.87106	2.05E-11	–	multiple drug resistance protein MarC
PA3489	-2.78195	0.000644	–	Na(+)-translocating NADH-quinone reductase subunit E
PA2604	-2.73994	1.02E-08	–	hypothetical protein
PA0890	-2.66271	1.04E-07	<i>aotM</i>	arginine/ornithine transporter AotM
PA5504	-2.61934	0.000857	–	D-methionine ABC transporter
PA3191	-2.60578	0.000136	–	two-component sensor
PA3211	-2.60557	0.001151	–	ABC transporter permease
PA1449	-2.589	0.000294	<i>flhB</i>	flagellar biosynthesis protein FlhB
PA3884	-2.53839	3.88E-05	–	hypothetical protein
PA1158	-2.47781	0.001312	–	two-component sensor
PA3102	-2.46575	0.004995	<i>xcpS</i>	general secretion pathway protein F
PA0392	-2.40655	0.0011	–	hypothetical protein
PA5256	-2.38917	8.64E-07	<i>dsbH</i>	disulfide bond formation protein
PA5043	-2.38491	4.92E-05	<i>pilN</i>	type 4 fimbrial biogenesis protein PilN
PA1691	-2.34904	0.000369	<i>pscT</i>	translocation protein in type III secretion
PA0184	-2.34177	0.002138	–	ABC transporter ATP-binding protein
PA4903	-2.31684	8.63E-05	–	major facilitator superfamily (MFS) transporter
PA5042	-2.31445	0.00109	<i>pilO</i>	type 4 fimbrial biogenesis protein PilO
PA2389	-2.27523	4.17E-05	<i>pvdR</i>	protein PvdR
PA4423	-2.24433	0.000448	–	hypothetical protein
PA2278	-2.22646	0.001024	<i>arsB</i>	

Table 2 (continued)

Locus Tag	Fold change	p-value	Gene	Product
PA3963	-2.21921	0.001037	–	arsenite-antimonite efflux pump ArsB
PA4954	-2.15623	0.001061	motA	flagellar motor protein MotA
PA4332	-2.13806	3.63E-06	–	hypothetical protein
PA0753	-2.12508	0.000215	–	hypothetical protein
PA5258	-2.11919	0.00121	–	hypothetical protein
PA0374	-2.10928	0.008412	<i>ftsE</i>	cell division protein FtsE
PA3839	-2.00907	0.007016	–	sodium:sulfate symporter
PA2155	-2.00863	0.007949	–	cardiolipin synthase 2
PA3857	-2.00479	5.9E-05	<i>pcs</i>	phosphatidylcholine synthase
Extracellular				
PA1092	-3.22535	9.38E-07	<i>flfC</i>	flagellin type B
PA1706	-2.54211	1.6E-05	pcrV	type III secretion protein PcrV
PA4699	-2.52982	0.00094	–	hypothetical protein
PA4525	-2.23285	0.000179	<i>pilA</i>	type 4 fimbrial PilA
PA0888	-2.12216	0.001585	aotJ	arginine/ornithine binding protein AotJ
Periplasm				
PA0766	-5.29562	1.88E-09	<i>mucD</i>	serine protease MucD
PA1081	-3.3145	1.96E-05	flgF	flagellar basal body rod protein FlgF
PA1092	-3.22535	9.38E-07	<i>flfC</i>	flagellin type B
PA4423	-2.24433	0.000448	–	hypothetical protein
PA0888	-2.12216	0.001585	aotJ	arginine/ornithine binding protein AotJ
Cytoplasm				
PA5471	-24.4276	3.21E-25	–	hypothetical protein
PA0395	-11.8779	5.47E-16	<i>pilT</i>	twitching motility protein PilT
PA4547	-8.10481	4.26E-17	<i>pilR</i>	two-component response regulator PilR
PA1758	-5.13388	2.96E-18	<i>pabB</i>	para-aminobenzoate synthase component I
PA1374	-4.67596	2.15E-08	–	hypothetical protein
PA1097	-4.59319	5.4E-12	fleQ	transcriptional regulator FleQ
PA3824	-4.29828	1.7E-08	<i>queA</i>	S-adenosylmethionine-tRNA ribosyltransferase-isomerase
PA0396	-4.02965	2.18E-08	<i>pilU</i>	twitching motility protein PilU
PA5261	-3.92591	8.22E-09	<i>algR</i>	alginate biosynthesis regulatory protein AlgR
PA3192	-3.78213	2.56E-06	<i>glrR</i>	two-component response regulator GlrR
PA4941	-3.50556	2.91E-07	<i>hflC</i>	protease subunit HflC
PA2235	-3.29461	3.35E-05	<i>pslE</i>	protein PslE
PA2043	-3.27156	8.88E-05	–	hypothetical protein
PA1572	-3.21822	1.43E-06	–	hypothetical protein
PA2260	-3.13241	5.52E-05	–	hypothetical protein
PA1110	-3.05002	5.45E-06	–	hypothetical protein
PA4524	-3.0098	3.82E-05	<i>nadC</i>	nicotinate-nucleotide pyrophosphorylase
PA5488	-2.97215	3.64E-08	–	hypothetical protein
PA3295	-2.89537	1.31E-05	–	HIT family protein
PA1987	-2.86813	1.69E-05	<i>pqqC</i>	pyrroloquinoline quinone biosynthesis protein PqqC
PA4942	-2.80403	3.75E-05	<i>hflK</i>	protease subunit HflK
PA1455	-2.76169	3.83E-07	flfA	flagellar biosynthesis sigma factor
PA3702	-2.75103	0.000245	<i>wspR</i>	two-component response regulator
PA1127	-2.68943	0.000323	–	oxidoreductase
PA2050	-2.68555	0.000271	–	RNA polymerase sigma factor
PA0338	-2.66712	0.000282	–	hypothetical protein
PA3350	-2.6418	4.9E-05	–	flagellar basal body P-ring biosynthesis protein FlgA
PA3070	-2.52559	0.0002	–	hypothetical protein
PA5493	-2.51171	0.000288	<i>polA</i>	DNA polymerase I
PA4231	-2.49648	0.001317	<i>pchA</i>	salicylate biosynthesis isochorismate synthase
PA3341	-2.48126	0.00066	–	transcriptional regulator
PA2550	-2.42534	0.003606	–	acyl-CoA dehydrogenase
PA4798	-2.40225	0.000452	–	hypothetical protein
PA3094	-2.39923	6.39E-06	–	transcriptional regulator
PA3945	-2.3736	0.001944	–	hypothetical protein

(continued on next page)

Table 2 (continued)

Locus Tag	Fold change	p-value	Gene	Product
PA4191	-2.34741	0.007044	–	iron/ascorbate oxidoreductase
PA3723	-2.30336	0.000115	–	FMN oxidoreductase
PA2261	-2.29666	0.002515	–	2-ketoglucuronate kinase
PA5115	-2.27699	0.004989	–	hypothetical protein
PA2255	-2.2466	0.000223	<i>pvcB</i>	paerucumarin biosynthesis protein PvcB
PA3444	-2.21674	0.003829	–	alkanesulfonate monooxygenase
PA1604	-2.21501	0.001079	–	hypothetical protein
PA3539	-2.18164	2.68E-06	–	hypothetical protein
PA3133	-2.17678	0.000596	–	transcriptional regulator
PA2254	-2.17364	0.002019	<i>pvcA</i>	paerucumarin biosynthesis protein PvcA
PA4902	-2.15901	0.001438	–	transcriptional regulator
PA2085	-2.12973	0.004435	–	ring-hydroxylating dioxygenase small subunit
PA2312	-2.1194	0.007103	–	transcriptional regulator
PA1950	-2.11364	0.010999	<i>rbsK</i>	Ribokinase
PA0413	-2.10034	1.71E-06	<i>chpA</i>	chemotactic signal transduction system protein
PA2496	-2.10004	0.000213	–	hypothetical protein
PA0937	-2.08702	0.000629	–	hypothetical protein
PA4508	-2.07877	0.002013	–	transcriptional regulator
PA3276	-2.07523	0.00202	–	hypothetical protein
PA4645	-2.06194	0.007127	–	hypoxanthine-guanine phosphoribosyltransferase
PA0893	-2.05346	0.005301	<i>argR</i>	transcriptional regulator ArgR
PA4617	-2.02286	0.003027	–	hypothetical protein
PA1186	-2.02149	0.001298	–	hypothetical protein
PA2003	-2.01423	0.008875	<i>bdhA</i>	3-hydroxybutyrate dehydrogenase
PA2120	-2.0132	0.003474	–	hypothetical protein
PA5429	-2.00956	0.001459	<i>aspA</i>	aspartate ammonia-lyase
Unknown				
PA0271	-8.13811	2.14E-06	–	hypothetical protein
PA3036	-3.92635	9.6E-08	–	hypothetical protein
PA4879	-3.67382	2.31E-15	–	hypothetical protein
PA2559	-3.55392	2.38E-09	–	hypothetical protein
PA3632	-3.35329	1.46E-05	–	hypothetical protein
PA4877	-3.24691	8.73E-05	–	hypothetical protein
PA5122	-3.17151	1.62E-05	–	hypothetical protein
PA2772a	-3.09495	8.4E-05	–	hypothetical protein
PA2009	-2.97886	2.88E-08	<i>hmgA</i>	homogentisate 1,2-dioxygenase
PA2238	-2.88138	5.37E-07	<i>pslH</i>	protein PslH
PA4949	-2.82889	0.000244	–	hypothetical protein
PA3663	-2.71457	0.001488	–	hypothetical protein
PA1842	-2.62859	0.000246	–	hypothetical protein
PA3762	-2.61944	0.000543	–	hypothetical protein
PA4323	-2.53315	0.000169	–	hypothetical protein
PA1740	-2.52962	5.72E-05	–	hypothetical protein
PA3084	-2.25805	0.001504	–	hypothetical protein
PA3064	-2.25203	1.25E-07	<i>pelA</i>	PelA protein
PA4882	-2.24131	0.000509	–	hypothetical protein
PA4958	-2.22137	0.001146	–	hypothetical protein
PA1031	-2.18395	0.005142	–	hypothetical protein
PA5182	-2.12966	0.000259	–	hypothetical protein
PA2454	-2.03603	0.002409	–	hypothetical protein
PA3493	-2.02273	0.008421	–	hypothetical protein
PA5306	-2.0162	0.003349	–	hypothetical protein

Bold are common negatively selected genes in both conditions of infection.

active release. The type IV pili are significant for host cell lysis or during the active release from the host. Skurnik et al. (2013) demonstrated that the Type IVa pili are critical for systemic dissemination in a murine model which substantiates our result. Interestingly, we observed the requirement of three proteases during cardiomyocyte infection without antibiotics. *mucD*, *hflK*, and *hflC* are the proteases conditionally essential during cardiomyocyte infection which localize in OMVs. MucD is a serine protease and an important virulence factor of *Pseudomonas* in plant, insect, nematode and mammalian infections. It is associated with the production of an extracellular toxin to kill *C. elegans* (Yorgey et al., 2001). Okuda et al., (2011) demonstrated that the MucD is required by

the toxin ExoS for IL-8 degradation to escape from the phagocytic killing by *P. aeruginosa*. Mochizuki et al. (2014) investigated the role of MucD in *P. aeruginosa* during corneal infection. MucD suppresses the IL-1 β , keratinocyte-derived cytokine and macrophage inflammatory protein-2 (MIP-2) and hence inhibits the neutrophil recruitment in the cornea during early infection. This facilitates *P. aeruginosa* to establish the infection by immune response evasion (Mochizuki et al., 2014). *hflKC* is a two-gene member operon that encodes the accessory factors for FtsH protease. FtsH belongs to ATP-dependent zinc metalloprotease which is the major determinant of aminoglycoside resistance. HflK and HflC act as the regulators of FtsH and aids in the intrinsic aminoglycoside resistance of *P. aeruginosa* (Hinz et al., 2011). Besides Aer, PonA, WbpJ, FlgE, and AotJ the other OMV localizing conditionally essential gene products are FtsX, PslE, PilA, AspA, PA4639, PA2800, PA4423, and PA5258. ArgR, FleQ, PA3341, PA3094, PA3133, PA4902, PA2312, and PA4508 are the conditionally essential transcriptional regulators during cardiomyocyte infection without amikacin.

3.2.3. Conditionally enriched *nuo* operon

Interestingly, a thirteen gene member operon that involves oxidative phosphorylation was found to be enriched during d-H9C2 infection. In both conditions of cardiomyocyte infection, most of the genes in *nuo* operon showed enhanced fitness. Table 3 provides the list of genes enriched in this operon with its fold change. Genes belong to this operon encode the subunits of type I NADH dehydrogenase which involves electron transfer to ubiquinone across the membrane by translocating the protons. Complex I play an important role in energy production. All the genes in this operon are essential for the production and assembly of the functional complex (Erhardt et al., 2012). An earlier study demonstrated the inability of *nuo* mutants to grow anaerobically in *P. aeruginosa* (Filiatrault et al., 2006). Velmurugan et al. (2007) revealed that the complex I is required for the inhibition of apoptosis during *Mycobacterium* infection. Particularly *nuoG* is the virulence gene that acts as an antiapoptotic factor. Our data shows the increased survival of the *nuo* mutants during cardiomyocyte infection which may due to the

Table 3

List of gene members and their fold change in an enriched operon.

Locus Tag	Fold change		Gene	Product
	With amikacin	Without amikacin		
PA2637	NA	31.02858694	<i>nuoA</i>	NADH dehydrogenase subunit A
PA2638	7.99793686	11.70917298	<i>nuoB</i>	NADH dehydrogenase subunit B
PA2639	37.19144061	13.30317973	<i>nuoD</i>	bifunctional NADH: ubiquinone oxidoreductase subunit C/D
PA2640	3.160018015	11.12431574	<i>nuoE</i>	NADH dehydrogenase subunit E
PA2641	1.90736	11.65111592	<i>nuoF</i>	NADH dehydrogenase I subunit F
PA2642	3.105325153	14.98323237	<i>nuoG</i>	NADH dehydrogenase subunit G
PA2643	8.758440298	17.20452316	<i>nuoH</i>	NADH dehydrogenase subunit H
PA2644	23.12150498	12.82784012	<i>nuoI</i>	NADH dehydrogenase subunit I
PA2645	NA	15.56377344	<i>nuoJ</i>	NADH dehydrogenase subunit J
PA2646	NA	19.71054376	<i>nuoK</i>	NADH dehydrogenase subunit K
PA2647	47.59193344	8.956379216	<i>nuoL</i>	NADH dehydrogenase subunit L
PA2648	3.266331948	7.326796663	<i>nuoM</i>	NADH dehydrogenase subunit M
PA2649	12.42174022	6.606816739	<i>nuoN</i>	NADH dehydrogenase subunit N

increased ROS tolerance (Teppo et al., 2018). Since the *nuo* operon plays a crucial role in energy production, increased survival of its mutant during infection is a captivating outcome that needs further analysis.

3.3. Live-dead assay and CFU enumeration

Mutants of PA1561 (*aer*), PA5205 and PA2639 (*nuoD*) were procured to validate the sequencing data. Fig. 4a shows the representative images of the live-dead assay. In comparison with wild-type, PA1561::lacZ and PA5205::phoA showed lesser cytotoxicity (Fig. 4b). PA2639::phoA showed significantly higher cytotoxicity in contrast with the wild-type. We found a significant ($p < 0.01$) reduction in the number of CFU in both PA1561::lacZ ($1E^{+04}$) and PA5205::phoA ($5.7E^{+05}$) (Fig. 4c). PA2639::phoA showed a significant ($p < 0.02$) increased CFU/ml ($2.35E^{+08}$) in contrast with PAO1 ($1E^{+07}$). One thousand-fold decrease of *aer* mutant and 100-fold decrease of PA5205::phoA in the intracellular load substantiates their conditional essentiality during cardiomyocyte infection. Ten-fold increased colonization of PA2639::phoA further validates the enhanced fitness of this gene after the disruption. PA2639 encodes NuoD- subunit of NADH: ubiquinone oxidoreductase. It appears that the cytotoxic effect of the mutants is related with their load during cardiomyocyte infection. Hence, the mutant showed enhanced fitness during infection resulted in severe cardiomyocyte death and vice versa.

4. Conclusion

The high incidence of *P. aeruginosa* especially in hospitals is due to the poor health status of the patients, the high prevalence of multidrug-resistant strains in hospitals, and the needless usage of broad-spectrum antibiotics. However, many efforts were taken in understanding the pathogenesis of *Pseudomonas* infection, better knowledge is obligatory for potential therapeutics to overcome the clinical burden due to *Pseudomonas*. Our findings aid in signifying the genetic requirements of *P. aeruginosa* during infection particularly at the intracellular-level. The core set of essential genes comprises 89 genes which are reported to be essential in all the earlier data-sets compared with this study. Severe reduction in the fitness of the *aer* mutant (−63.381 fold) and the reduced cytotoxicity (Fig. 4) suggest the importance of aerotaxis during intracellular survival. Further investigation of the core set essential genes and conditional essential genes provide promising drug targets for this ubiquitous organism. Specifically, genes associated with flagella and T3SS are important for adhesion and invasion, while pili and proteases play a crucial role during host cell lysis. Garai et al. (2019) also showed the importance of ExoS effector in promoting the phagosomal escape and macrophage lysis during intramacrophage life of *P. aeruginosa*. Besides the genes encode hypothetical proteins, genes associated with transcription, amino acid transport & metabolism, and signal transduction play an important role during stringent intracellular lifestyle, whereas genes related to motility are important in the extracellular

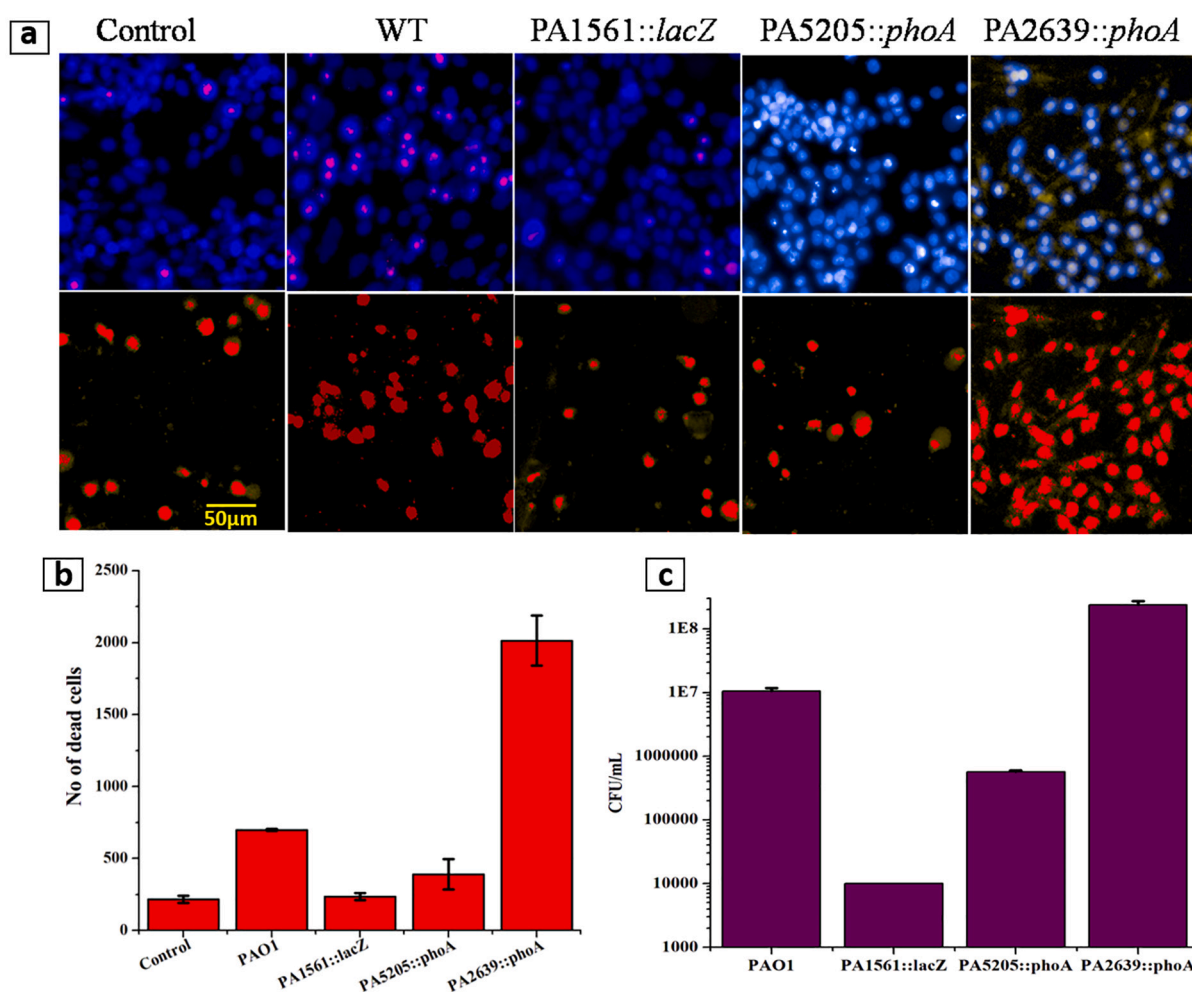


Fig. 4. a. Representative images of the infected d-H9C2 cells with wild-type and mutants. b. The number of dead cells was quantified at 12 h of post-infection using PI stain. PI-stained cells were quantified using Harmony 3.0 software. c. CFU enumerations: d-H9C2 cells were infected with the mutants, and CFU was enumerated after 12 h of infection. The mean values of three independent experiments were shown in the bar chart.

mode. OMVs play a significant role in the progression of the infection. Interestingly, the mutants of *nuo* operon are enriched in both conditions of infection and further study is needed to understand the molecular insights of their role in infection.

Conflicts of interest

The authors declare no conflict of interest.

CRediT authorship contribution statement

Jothi Ranjani: Conceptualization, Investigation, Visualization, Formal analysis, Writing – original draft, Writing – review & editing. **Ramamoorthy Sivakumar:** Conceptualization, Validation, Writing – review & editing. **Paramasamy Gunasekaran:** Validation, Supervision, Resources, Writing – review & editing. **Ganesan Velmurugan:** Conceptualization, Writing – review & editing. **Subbiah Ramasamy:** Conceptualization, Supervision. **Jeyaprakash Rajendhran:** Conceptualization, Methodology, Supervision, Funding acquisition, Writing – review & editing.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.meegid.2022.105231>.

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